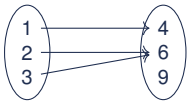
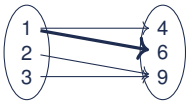
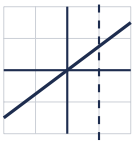
Functions

NC.8.F.1 • NC.8.F.3 • NC.8.F.4

Unit 6 • Front

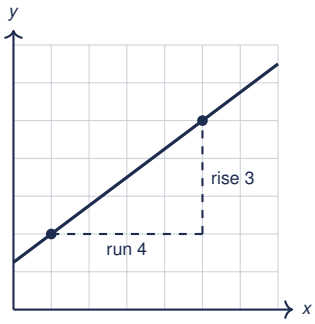
1 What Is a Function?

Function: each input (x) has **exactly one** output (y). An x can **not** repeat with a different y.

	FUNCTION ✓	**NOT a function ×**
Ordered pairs	$(1,4),\ (2,6),\ (3,6)$ Outputs can repeat. Inputs can't.	$(1,4),\ (\mathbf{1},7),\ (3,6)$ Input 1 has two outputs.
Mapping	 One arrow leaves each input.	 Input 1 has two arrows.
Graph	 **Vertical line test:** hits once.	 Hits twice: one x, two y's.

2 Slope = Rate of Change

$$m = \frac{\text{change in } y}{\text{change in } x} = \frac{\text{rise}}{\text{run}}$$



$$m = \frac{3}{4}$$

Two points: $(2,3)$ and $(6,11)$

$$m = \frac{11-3}{6-2} = \frac{8}{4} = 2$$

4 kinds of slope:



up, down, flat, straight up

Table: track the change.

Δx	x	y	Δy	$\frac{\Delta y}{\Delta x}$
	0	1		
+2	2	7	+6	3
+3	5	16	+9	3
+1	6	19	+3	3

x jumps are **uneven**? Divide each one!
Same every time → **linear**, $m = 3$.

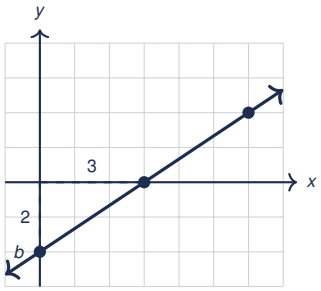
3 Slope-Intercept Form: $y = mx + b$

$$y = \mathbf{m}x + \mathbf{b}$$

m = **slope** (rate of change)
b = **y-intercept** (starting value, where $x = 0$)

Graph $y = \frac{2}{3}x - 2$

1. Start at b: plot $(0, -2)$.
2. Use m: up 2, right 3. Plot.
3. Repeat. Draw the line.



From a story: A taxi charges \$3 to start plus \$2 per mile.

start → $b = 3$ per mile → $m = 2$

$\boxed{y = 2x + 3}$

Linear or nonlinear?

Linear straight line
same rate of change
$y = mx + b$ (ex: $y = 4x - 1$)

Nonlinear curve or bends
rate changes
ex: $y = x^2$, $y = \frac{4}{x}$

Using Functions

NC.8.F.2 • NC.8.F.5 • NC.8.EE.8

4 Compare Two Functions

Function A

x	0	1	2
y	5	8	11

$m = 3$ (up 3 each time)
$b = 5$ (when $x = 0$)

Function B

$y = 4x + 2$

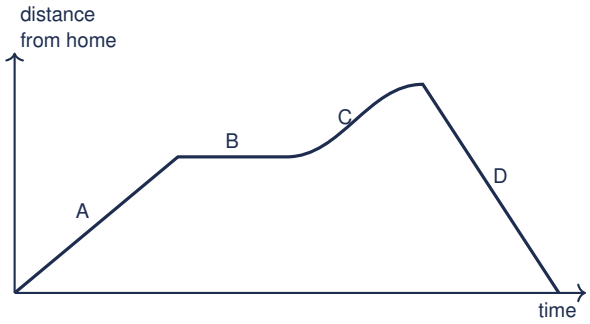
$m = 4$
$b = 2$

1. Find m and b for each one.
2. Compare.

Faster rate: B, because $4 > 3$.
Higher start: A, because $5 > 2$.

Works for tables, graphs, equations, and stories.

5 Qualitative Graphs: Tell the Story



A	increasing, linear	walks away at a steady speed
B	constant (flat)	stops
C	increasing, nonlinear	speeds up
D	decreasing, linear	goes back home

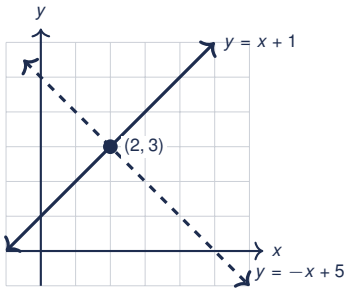
Read left to right. Up = increasing. Down = decreasing. Flat = constant.

6 Systems of Equations

Solution = the point where the lines **cross**. It works in **both** equations.

Graphing

$y = x + 1$ $y = -x + 5$



Substitution: set them equal.

$$\begin{aligned} x + 1 &= -x + 5 \\ \underline{+x} & \quad \underline{+x} \\ 2x + 1 &= 5 \\ \underline{-1} & \quad \underline{-1} \\ 2x &= 4 \\ \frac{2x}{2} &= \frac{4}{2} \\ x &= 2 \end{aligned}$$

Then $y = 2 + 1 = 3$. $\boxed{(2, 3)}$

Inspection: compare m and b.

Different m	**one** solution lines cross
Same m, **different** b	**no** solution parallel lines
Same m, **same** b	**infinitely many** same line

Ex: $y = 2x + 1$ and $y = 2x - 4$
Same m, different $b \rightarrow$ **no solution**

Check $(2, 3)$: $y = x + 1 \rightarrow 3 = 2 + 1$ ✓ $y = -x + 5 \rightarrow 3 = -2 + 5$ ✓ Works in both. Done!

7 Write the Equation, Then Explain It

From two points: $(2, 7)$ and $(4, 11)$

1. Slope: $m = \frac{11 - 7}{4 - 2} = \frac{4}{2} = 2$
2. Find b: put one point into $y = mx + b$.

$$\begin{aligned} 7 &= 2(2) + b \\ 7 &= 4 + b \\ \underline{-4} & \quad \underline{-4} \\ 3 &= b \end{aligned}$$

3. Write it: $\boxed{y = 2x + 3}$

Explain m and b in the story

Savings: $y = 15x + 40$
x = weeks y = dollars saved

$m = 15$ **Rate:** saves \$15 **per week**

$b = 40$ **Start:** had \$40 at **week 0**

Predict: after 6 weeks,
$y = 15(6) + 40 = 90 + 40 = \$\mathbf{130}$

Key words: **per, each, every** $\rightarrow$ slope. **start, fee, initial** $\rightarrow$ y-intercept.